



COLLEGE OF COMPUTER STUDIES
Information Technology Department

COURSE SYLLABUS

Ng, Fitzgerald M.

COURSE TITLE	Algorithms and Complexity	
COURSE CODE	AL101	
CREDIT UNITS	3 units	
COURSE PREREQUISITE	MS101	
COURSE DESCRIPTION	The Algorithms and Complexity focuses on the mathematical modelling and analysis of algorithmic processes. Logic is emphasized in this course, and topics include proof and theory (including inductive and deductive proofs), propositional and predicate logic, set theory, algorithms (including recursion), trees, relations and functions, counting and probability, and elements of the theory of directed and undirected graphs (including Dijkstra's shortest path algorithm). Additionally, an introduction to complexity of algorithms and recurrence relations are included. Upon completion, students will be able to solve real world problems and use appropriate models for analysis.	
VISION		CORE VALUES
To be recognized as the #1 local university of employable graduates.		Jointness of undertaking and Organizational adaptability with Yoke of Efficiency and Effectiveness
MISSION		
To provide a comprehensive education that enhances the lives of QCU students for nation building and as world citizens.		
INSTITUTIONAL OUTCOMES		COURSE OUTCOMES
QCU graduates are: Qualified in their chosen field locally and globally Competent and effective communicators Unique in demonstrating their sense of genuine nationalism and patriotism		At the end of the course, the student should be able to: LO1. Use systems thinking to analyze business processes and identify problems and opportunities that can be solved and supported by technology.
PROGRAM OUTCOMES		
Common to all Programs The graduates of BSIT have the ability to a.) Articulate and discuss the latest developments in the specific field of practice; b.) Effectively communicate orally and in writing using both English		



and Filipino

- c.) Work effectively and independently in multi-disciplinary and multi-cultural teams
- d.) Act in recognition of professional, social, and ethical responsibility
- e.) Preserve and promote "Filipino historical and cultural heritage"

Common to the discipline

The graduates of BSIT must have the ability to

- a.) Analyze complex problems, and identify and define the computing requirements needed to design an appropriate solution
- b.) Apply computing and other knowledge domains to address real word problems
- c.) Design and develop computing solutions using a system-level perspective
- d.) Utilize modern computing tools

LO2. Apply appropriate tools, methods, models/techniques in systems analysis and design.

LO3. Develop and defend a project design proposal to different audiences.

OUTCOMES-BASED TEACHING AND LEARNING (OBTL) PLAN

WEEK	LEARNING OUTCOMES	TOPIC	METHODOLOGY <i>Flexible Teaching Learning Activities</i>	LEARNING AND TEACHING RESOURCES	ASSESSMENT and TASKS
1	- Know the importance of the subject. - Value the QCU Mission and Vision. - Determine the course discussion and requirements. - Determine the Grading System of the University	Course Orientation And Course Introduction - Mission - Vision - Course Syllabus - Grading System - Classroom policies	Blended Learning (Online/Onsite) Lecture Discussion	PowerPoint Presentation Google Classroom Google Meet	Oral Recitation/Selected Responses



2	● Define and understand the meaning of the algorithm. ● Identify and understand the different examples of algorithm ● Identify the kinds of problems solved by algorithms. ● Appreciate the importance of algorithmic p.	Introduction To Algorithms ● Definition of Algorithm ● Examples of Algorithm ● Fundamentals of Algorithmic Problem Solving ● Kinds of Problems Solved by Algorithms	Blended Learning (Online/Onsite) Lecture Discussion	PowerPoint Presentation Google Classroom Google Meet	Oral Recitation Brainstorming Assignment 1 Activity
---	--	--	--	--	--



3	- Identify the classification and application of data structures. - Differentiate the different types of data structures. - Appreciate the application and importance of data structures.	Data Structures - Definition of Data Structures Types of Data Structures - Linear Data Structure - Graph Data Structure - Trees Data Structure	Blended Learning (Online/Onsite) Lecture Discussion	PowerPoint Presentation Google Classroom Google Meet	Oral Recitation Brainstorming Activity
4-5	- Define and understand analysis algorithm - Identify and understand the framework in analyzing the efficiency of an algorithm - Apply the different asymptotic notation in comparing the order of growth - Differentiate recursive and non-recursive	Analysis of the Efficiency of Algorithm - Definition of Analysis of Algorithm - Time and Space Efficiency - Empirical Analysis of Algorithm - Order of Growth - Algorithm Case Efficiency - Asymptotic Notation - Big-Oh Notation - Big-Omega Notation - Big-Theta Notation - Basic Efficiency Classes - Mathematical Analysis of Recursive Algorithm	Blended Learning (Online/Onsite) Lecture Discussion	PowerPoint Presentation Google Classroom Google Meet	Oral Recitation QUIZ 1 Activity



	algorithm	Mathematical Analysis of Non-recursive Algorithm			
6-7	- Identify the brute force algorithm design technique - Appreciate the importance of brute force algorithm design technique - Apply brute force algorithm design technique in problem-solving	BRUTE FORCE ALGORITHM DESIGN TECHNIQUE - Definition of Brute Force Algorithm Design Technique - Importance of Brute Force Algorithm Design Technique - Application of Brute Force Algorithm Design Technique in the following: - Selection sort - Bubble sort - Sequential search - String matching - Exhaustive search	Blended Learning (Online/Onsite) Lecture Discussion	PowerPoint Presentation Google Classroom Google Meet	Oral Recitation Assignment 2 Activity



	● Apply the different variations of the decrease-and-conquer algorithm design technique	● Application of Decrease-and-Conquer Algorithm Technique: ● Insertion Sort ● Depth-first Search (DFS) ● Breadth-first Search (BFS) ● Graph Traversal			
11	● Recognize and describe Transform-and-Conquer Algorithm Design Technique ● Use the application of transform-and-conquer algorithm design technique to problem-solving ● Appreciate the application of the transform-and-conquer algorithm design technique to problem-solving	Transform-and-Conquer Algorithm Design Technique ● Definition of Transform-and-Conquer Algorithm Design Technique ● Application of the Transform-and-Conquer Algorithm Design Technique: ● Presorting Algorithm ● Gaussian Elimination ● Balanced Search Tree ● Problem Reduction	Blended Learning (Online/Onsite) Lecture Discussion	PowerPoint Presentation Google Classroom Google Meet	Oral Recitation Activity QUIZ 1



12-13	• Comprehend the Greedy Algorithm technique • Describe the Greedy Algorithm design technique and identify its importance • Apply the different types of Greedy Algorithm techniques in solving the problem	Greedy Algorithm • Definition of Greedy Algorithm Design Technique • Importance of Greedy Algorithm Design • Technique Types of Greedy Algorithm • Prim's Algorithm • Kruskal's Algorithm • Dijkstra's Algorithm • Huffman Tree	Blended Learning (Online/Onsite) Lecture Discussion	PowerPoint Presentation Google Classroom Google Meet	Oral Recitation Activity Assignment 2
14	• Synthesize the dynamic programming algorithm technique • Describe the elements of dynamic programming algorithm technique	Dynamic Programming • Applying dynamic programming • Elements of dynamic programming	Blended Learning (Online/Onsite) Lecture Demonstration	PowerPoint Presentation Google Classroom Google Meet	Oral Recitation QUIZ 2 Activity



QUEZON CITY UNIVERSITY

SAN BARTOLOME | SAN FRANCISCO | BATASAN HILLS



15	● Identify the different types of graph algorithm technique and their uses ● Synthesize the application of graph algorithm technique	Graph Algorithm ● Graphs and Games ● Traversing Trees ● Depth-first Search: Undirected graphs ● Depth-first Search: Directed graphs ● Breadth-first Search ● Backtracking ● Branch-and bound ● Minimax Principle	Blended Learning (Online/Onsite) Lecture Discussion	PowerPoint Presentation Google Classroom Google Meet	Oral Recitation Activity
16-17	● Present a case study -- Application - Steps to Improve - Proposal and Recommendation	Presentation Presentation of the simulation of the proposed algorithm.	Blended Learning (Online/Onsite) Discussion	Google Classroom Google Meet	Facilitation of assigned topics
18	FINAL EXAMINATION				



REFERENCES:

- Cormen T., Leiserson C., Rivest R., & Stein C. (2009). Introduction to Algorithms Third Edition. The MIT Press, Cambridge, Massachusetts
- Fleck, Margaret M., (2013). Building Blocks for Theoretical Computer Science. Version 1.3 (January 2013)
- Lehman, Eric F., Leighton, Thomson & Meyer, Albert R. (2018). Mathematics for Computer Science. June 2018

SUGGESTED READINGS AND REFERENCES

OTHERS / ONLINE REFERENCES

- <https://www.geeksforgeeks.org/>
- <http://www.freebookcentre.net/CompuScience/free-computer-algorithm-books.html>

COURSE POLICIES AND STANDARDS

Expectation from Student

The student's responsibility is to come to each class prepared, and to participate actively in the discussions. Student should read the assigned coursework prior to class, and to present reports as scheduled. S/he is also expected to take all quizzes and examinations and to submit all course requirements on the dates scheduled. S/he is expected to observe proper and polite behavior at all times, adhering to the rules and policies of QCU.

Language of Instruction

The language of instruction is English.

Attendance

Students are required to attend all classes starting with the first meeting. Non-attendance in any required class or academic activity constitutes an absence. A student who has been absent for more than 20 percent of the hours of recitation, lectures, or any other scheduled work in this course shall be automatically dropped from the class roll and the Registrar shall be advised accordingly.

Student Rights and Responsibilities

The QCU Student Manual establishes students' freedoms and protections as well as expectations of appropriate behavior and ethical academic work. The Handbook includes items such as the Policy on Student Rights, the Policy on Student Conduct, and the Academic Integrity Policy.

Academic Integrity

Any work that students submit should be their own work (i.e., not borrowed/copied from any other source, including their classmates). When using other people's ideas to substantiate their ideas, students are expected to properly cite the original source. Proper citation procedures shall be discussed in class. Any act of cheating or plagiarism shall be treated in accordance with the Institute's Policy on Academic Integrity. Depending upon the individual violation, students could face penalties ranging from failing the assignment to failing the class.



QUEZON CITY UNIVERSITY

SAN BARTOLOME | SAN FRANCISCO | BATASAN HILLS



Grading System

The College of Computer Studies follows the university grading system given below:

Middle term Grade (MTG)

*Class Standing	60%
Midterm Exam Grade	40%
Total	100%

Final Term Grade (FTG)

*Class Standing	60%
Final Exam Grade	40%
Total	100%

FINAL GRADE:

Midterm Grade	50%
Final Term Grade	50%
Total	100% or (MTG + FTG)/2

*Class Standing (60%)

Online Discussion/ Presentation	5%
Assignment	10%
Quizzes	25%
Output/Machine Problem	20%

Rubrics for Course Output/Requirements

Group Project Criteria:

Peer Evaluation	-	20%
Group Grade (Presentation)	-	50%
Individual grade	-	30%
		100%

Student Course Portfolio

All exercises, activities and requirements for the course must be compiled by the students and must be submitted to the instructor before the end of the semester.

Learners with Disabilities

The University and the College of Computer Studies strives to make all learning experiences as accessible as possible. If a student anticipates or experiences barriers based on their disability (including mental health, chronic or temporary medical conditions), the students need to inform their instructor immediately so that they can discuss options privately. To establish reasonable accommodations, the



QUEZON CITY UNIVERSITY

SAN BARTOLOME | SAN FRANCISCO | BATASAN HILLS



students need to register in the "List of Learners with Disabilities" at the Office of Student Affairs and Services.

Syllabus Flexibility

The instructor reserves the right to amend or change this syllabus as needed. For example, the instructor may modify content, and/or substitute assignments in response to institutional, weather, or class situations. Any modification, however, must be communicated by the instructor to the Dean through the Chairperson. Approved changes shall be discussed by the instructor in class.

EFFECTIVITY AND REVISION		PREPARED BY	REVIEWED BY	RECOMMENDING APPROVAL	APPROVED BY
Issue No. 02	Revision No. 02	 Redentor G. Bucaling Jr. Faculty	 Maria Aura Impang, LPT, MSIT Program Chair, Information Technology Department	 Dr. Isagani M. Tano Dean, College of Computer Studies	 Dr. Randolph Antonio Martinez Vice President for Academic Affairs
QA Doc. No.	Effectivity AY 2022-2023				

The original copy of this document is filed at the Office of the Dean copy furnished the Quality Assurance Office.
//gar22

A-24
JAN 1 2023 LPT, MSIT